CV-CIM: A Hybrid Domain Xor-Derived Similarity-Aware Computation-in-Memory Supporting Cost–Volume Construction

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***Abstract*— Cost–volume construction, a cornerstone of stereo vision processing, has been widely utilized across diverse domains such as robotics, autonomous vehicles, 3-D reconstruction, and augmented/virtual reality applications. However, this process necessitates substantial parameter sizes and continuous memory data accesses, imposing considerable demands on memory bandwidth. Computation-in-memory (CIM) emerges as a promising technology to alleviate this memory bottleneck by its ability to execute parallel computations within structured arrays. Nevertheless, the direct deployment of cost–volume construction on CIM platforms faces several challenges, including functional limitations, reduced compute margins, and structural rigidity. To overcome these obstacles, we present a novel CIM architecture, named cost–volume CIM (CV-CIM), which incorporates the following key features. First, the XOR-derived logic integrates fundamental arithmetic functions and heterogeneous CIM array (CA) function together to complete requisite cost computation. Second, the pixel-wise similarity is discovered to enlarge the bit sparsity. Then, sparse-mode CIM leverages the resulting sparsity to enlarge the XOR-based computing margin. Third, dual degrees of freedom peripherals collectively contribute to latency reduction through dynamic row computation skipping (row peripheral) and the overlapping of data loading and computing (column peripheral). A test chip is fabricated in 28-nm technology and operates within a voltage range of 0.6–0.9 V and a frequency range of 50–286 MHz. Notably, CV-CIM demonstrates support for L0/L1/L2 cost computations and achieves a remarkable peak energy efficiency of 1158 TOPs/W.**

***Index Terms*— Artificial intelligence, computation-in-memory (CIM), cost–volume, heterogeneous architecture, mixed signal.**

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I. Introduction

STEREO vision has been applied to a multitude of mobile applications, including robotics, autonomous vehicles, drones, and augmented reality/virtual reality (AR/VR) [1], [2], [3]. Cost–volume construction, a critical step in stereo vision, establishes pixel-wise correspondences between images from multiple perspectives, ultimately leading to depth information extraction. The subsequent extraction of depth information relies on the geometric relationships between these matched pairs. However, the stringent real-time requirements (30 frames/s) of image pair streams, coupled with data reloading, necessitate continuous data access during cost–volume construction. This, in turn, imposes a considerable demand on memory bandwidth (>0.25 Tb/s), which mismatches with the memory constraint and battery limitation of the edge device. The prior digital accelerator has demonstrated that power consumed by memory access is 3.06× higher than cost computing (31.2 versus 10.2 mW) [4]. Computation-in-memory (CIM) has emerged as a promising paradigm to address the memory bottleneck inherent in von Neumann architectures, which facilitates processing directly within the physical structure of the storage array. In light of the memory size limitations of edge devices, numerous studies have explored static random access memory (SRAM)-based CIM architectures, demonstrating their potential for competitive energy efficiency [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16]. However, the direct deployment of existing CIM approaches to cost–volume construction encounters three primary challenges, collectively termed the "3R challenges."

1) The first challenge is the **restricted** CIM logic, which stems from the limited range of computational functions that conventional CIM architectures can support. Cost–volume construction requires flexibility in selecting appropriate cost functions (e.g., Hamming distance [17] and Manhattan distance [18]) based on specific image properties and scenarios. Different distance algorithms cover multiple arithmetic logic [19], [20], [21], [22]. However, traditional CIM architectures are often limited to multiply and accumulation (MAC) operations, rendering them incapable of directly executing the intricate logic required by cost–volume functions.

2) The second challenge is the **reduced** computing margin. The computing margin, ideally defined as the ratio of the available physical quantity range to the accumulated output range, influences the quantization accuracy and the system's tolerance for errors. As output values increase, the voltage margin corresponding to each value diminishes, potentially degrading quantization accuracy. Cost–volume construction, due to its sensitivity to pixel-wise deviations, is particularly susceptible to the reduced computing margin.
3) The third challenge is the **rigid** structured array. The CIM arrays (CAs) typically possess fixed row and column activation patterns, impeding the efficient execution of algorithms with dynamic dataflows, such as cost–volume construction with its sliding target pixel and search region. The array is unable to accommodate the alternating data during the sliding process. Hence, it takes multiple cycles to process a single search region, leading to low CIM utilization. Furthermore, the sliding nature of the algorithm exacerbates this challenge by introducing additional data updates, stalling computations, and compromising system performance.

To address the aforementioned challenges, this work presents a novel CIM architecture, named cost–volume CIM (CV-CIM). It incorporates several innovative features to overcome the limitations of conventional CIMs.

1) XOR*-Derived Logic for Enhancing Flexibility:* CV-CIM leverages an XOR-derived logic scheme that integrates multiple arithmetic functions essential for cost–volume construction. This approach considers the area overhead associated with digital CIM and the functional limitations of analog CIM by employing hybrid XOR-CIMs to compute cost functions.
2) *Sparse-Mode CIM for Enlarging Margin:* CV-CIM exploits pixel-wise similarities to amplify computing margin. This is achieved through a sparse-mode CIM that transforms initial data into the differences between neighboring pixels. Differential computing inherently induces bit-wise sparsity, then lowers the accumulation value, and enhances the margin.
3) *Dual-Degree-of-Freedom Peripherals for Optimizing Utilization:* Column-wise circuitry unlocks idle columns, enabling the parallelization of computing and data loading. Concurrently, the row-wise peripheral employs an epipolar-first mechanism to selectively skip redundant search operations and dynamically activate rows to adapt to shifted regions.

By integrating three features, CV-CIM achieves 453∼1158-TOPs/W energy efficiency and 13.1∼27.1-TOPs/mm^2 area efficiency on evaluated benchmarks, which outperforms 1.06∼5.29× and 1.35∼5.01× than state-of-the-art (SOTA) CIMs, respectively.

This work is an extension of our ISSCC 2023 work [23], providing a deeper understanding of CV-CIM's design, capabilities, and potential impact. The rest of this article is organized as follows. Section II provides background and motivation. Section III presents the overall architecture. Sections IV–VI details three features. Section VII introduces the chip prototype. Section VIII shows the experiment results and Section IX concludes this work.

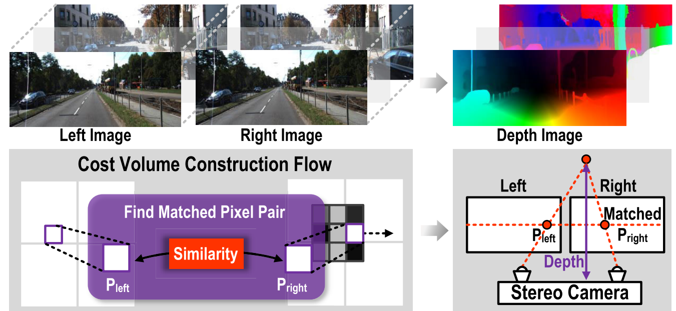

Fig. 1. Workflow of cost–volume construction, from image pair to disparity.

TABLE I
COST–VOLUME CONSTRUCTION ACCURACY AND LATENCY

Scenario		SAD [18]	Census [25]	Correlation [27]	Concate [28]
Well-posed[1]	Accuracy	85.2%	96.75%	97.6%	98.5%
	Latency	0.01s	0.07s	0.21s	0.17s
Ill-posed[2]	Accuracy	52.6%	66.11%	94.1%	95.6%
	Latency	0.01s	0.07s	0.21s	0.17s

[1] Non-occluded region with 5 pixels threshold.
[2] Reflective region with 5 pixels threshold.

II. BACKGROUND AND MOTIVATION

A. Cost–Volume Construction

In a manner analogous to human binocular vision, stereo cameras leverage a pair of images captured from two viewpoints to extrapolate depth information. Based on the principle of triangulation, the object distance can be computed from the disparity between the corresponding pixel projections in the two images [24]. As illustrated in Fig. 1, this disparity, characterized by a subtle horizontal offset between corresponding points (P_{left} and P_{right}), exhibits an inverse relationship with the distance. The core of the algorithm, therefore, involves the accurate identification of pixel correspondences between the two images. The cost–volume construction estimates the similarity, often referred to as matching cost, between a given target pixel in the left image and an array of candidate pixels in the right image. The pixel pair with the lowest matching cost is ultimately chosen as the final winner and is subsequently utilized to compute the distance [25].

The matching cost–volume is constructed by iterating exhaustively over the entire image pixel, thereby dominating the computing latency. Various algorithms have demonstrated distinct cost functions, such as sum of absolute difference (SAD) [18], census transform [17], correlation [21], and concatenation [26]. These functions fundamentally quantify the distance (L0/L1/L2) between the corresponding pixels, with varying degrees of adaptability to different imaging scenarios. Table I presents a comprehensive evaluation of multiple cost functions based on the KITTI 2012 benchmark. SAD is the oldest and simplest algorithm used in stereo vision, which adopts L1 distance (Manhattan distance) to calculate disparity. The census transform, which estimates L0 distance (Hamming distance), demonstrates remarkable accuracy (96.7%)

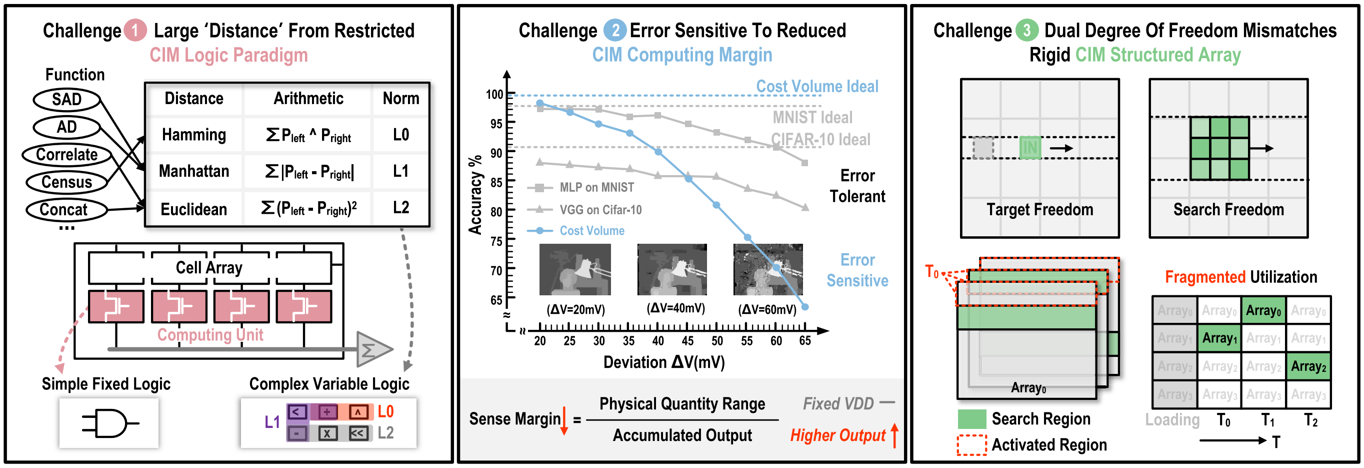

Fig. 2. 3R challenges of deploying CIM on cost–volume construction: restricted logic, reduced computing margin, and rigid CA.

and high frame rate in well-posed test cases within benchmarks. However, its performance deteriorates significantly (>30% accuracy loss) in scenarios such as occlusions, reflective regions, transparency, and repetitive patterns. To address these challenges, certain depth extraction networks have adopted the L2 correlation-based distance estimation, achieving a substantial accuracy improvement of 27.9% in ill-posed scenarios. The L2 distance (Euclidean distance) measures the spatial distance between two points. In stereo vision, the Euclidean distance is utilized to estimate multi-dimensional disparity between two pixels. However, this computational advantage comes at the cost of a 17.3× increase in latency. To accommodate the diverse range of conditions encountered in real-world applications, CV-CIM integrates variable distance functions, enabling adaptive selection based on image characteristics.

B. Challenges of Deploying Cost Construction on CIM

CIM is a promising architecture for resolving the memory bottleneck, which encodes digital values as physical quantities, such as current or charge. Computations are then executed within the memory array itself, harnessing physical phenomena such as Kirchhoff's laws or charge sharing. The computation is completed within the memory array to avoid energy-expensive data movement.

Prior CIM works have demonstrated attractive energy efficiency performance in multiple tasks, including binary neural networks (BNNs), convolution neural network, transformer model [11], [29], [30], [31], [32]. An SRAM-based Conv-RAM is utilized to accelerate CNN, which realizes multiply and average (MAV) operations based on 10T cell and local computing units [33]. An SRAM CIM performed MAC between 8-b weight and activation for deep neural network (DNN), which employs a compact 6T memory cell and avoids the read-disturb issue [5]. Moreover, compact 6T CIM architectures have been successfully implemented for the acceleration of lower precision BNNs [34]. Despite the impressive energy efficiency gained by reduced data movement in conventional CIM architectures, their deployment to cost–volume construction remains hindered by limitations arising at multiple levels—from cell logic to individual compute units and the entire array structure, as illustrated in Fig. 2.

1) Cost–volume construction employs diverse cost functions, each utilizing distinct distance metrics (e.g., Hamming distance in census transform and Manhattan distance in SAD). These commonly employed variable functions, encompassing L0, L1, and L2 distances, necessitate the integration of relatively complex logic, including operations such as XOR, multiplication, summation, and comparison. However, conventional CIMs exhibit a constrained function, predominantly restricted to multiplication and accumulation operations, which are optimized for matrix multiplication, hence limiting the array's capabilities to only "AND" logic.
2) The computing margin of CIM equals the feasible physical quantity range divided by the accumulated output range. In CIM with a fixed supply voltage (VDD), higher outputs lead to a reduction in this margin, making the quantized results susceptible to deviations caused by process, voltage, and temperature (PVT) variations. This poses a significant challenge for cost–volume construction, which is sensitive to errors due to its reliance on pixel-wise information for direct depth map generation. Compared with the ideal scenario, experiments have shown that under reduced compute margin conditions, accuracy degradation in cost–volume construction can exceed 35%, significantly surpassing 11% observed in neural network applications subjected to the same deviation.
3) Cost–volume construction iterates over entire images, with each target pixel associated with a variable search region. The data within these search regions are typically dispersed across the CA. Consequently, conventional CIM architectures, constrained by fixed activation patterns, require multiple cycles to engage all necessary data with limited utilization. For instance, when processing the census transform with a 16 × 16 search region, utilization is bounded to merely 23%. Furthermore, when the footprint of the sliding search region is larger

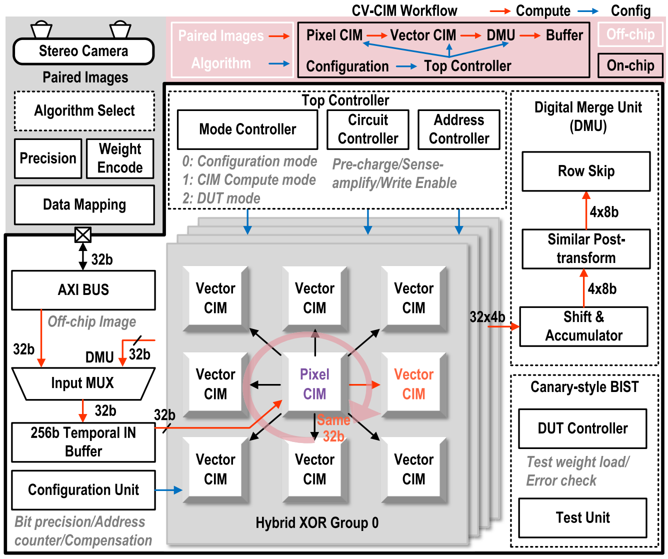

Fig. 3. Overall architecture and corresponding workflow of CV-CIM.

than the available array storage capacity, data loading is compulsory. Under a sequential update-compute approach with a 16 × 16 search region, 14% of the overall latency is attributable to data loading.

To reconcile the functional requirements and robustness demands of cost–volume construction, we propose a novel CV-CIM to support diverse distance functions, offer an enhanced compute margin, and accommodate variable search regions.

III. CV-CIM Overall Architecture

Fig. 3 shows the overall architecture of CV-CIM, including four hybrid XOR-CIM groups, a top-level controller, a digital merge unit (DMU), a temporal buffer, a canary-style built-in self-test (BIST) unit, and other peripherals. The hybrid XOR-CIM group is the fundamental component, each incorporating eight vector CIMs and a pixel CIM according to the operation analysis of the distance algorithm. The workflow is detailed in Fig. 3.

1) *Image Acquisition and Configuration:* The stereo camera captures paired images and the algorithm is decided off-chip based on the specific image characteristics. The designated algorithm, along with its associated data precision requirements, the number of iterations, and encoded data, is written into the configuration unit and on-chip buffer ahead of computing.
2) *On-Chip Data Processing:* Once the chip completes the configuration, the CAs enter the computing mode. The pixel CIM unit performs the initial processing of the input data, and the resulting outputs are shared across eight parallel vector CIMs for vector operations. The pixel/vector CA is configured with 16 × 64 cell and 32 XOR computing units activated simultaneously. The accumulated analog current results are quantized by a 4-bit flash analog-to-digital converter (ADC). The outputs from the ADCs of all vector CIMs (32 × 4 bits) are further processed within the DMU. This unit performs bit-wise shifting and accumulation operations to generate the final results. The DMU accumulates the partial results from each CIM based on the bit significance or CIM mode. It subsequently updates the temporal buffer with the merged results.
3) *Robustness Testing:* A canary-style BIST unit is embedded within the architecture, testing the resilience of the analog units under PVT variations.

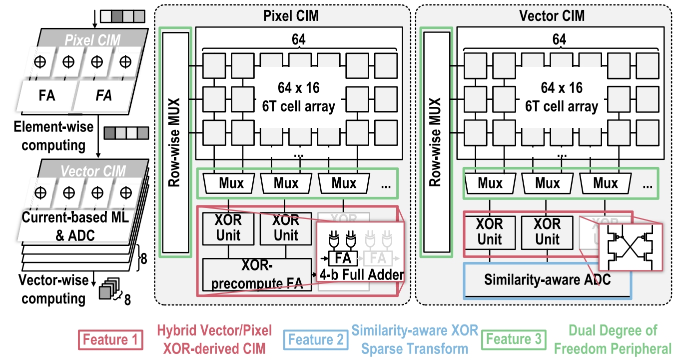

Fig. 4. Pixel/vector CIM and specific features.

Fig. 4 details the distinctions between vector CIM and pixel CIM. The pixel CIM executes pixel-wise (element-wise) computations via digital XOR logic, while area-optimized full adders (FAs) accumulate the results from two adjacent columns. This accumulated outcome is subsequently shared by eight vector CIMs, which employ current-based XOR to perform vector-wise computations. Results of the entire row are merged and quantized by a similarity-aware ADC. Fig. 4 also presents three features to tackle the limitations of prior CIMs. First, a hybrid XOR unit integrates various operations that are derived from XOR logic. The combination of vector- and pixel-wise operation supports different distance algorithms. Second, the similarity-aware transform is implemented to reduce the accumulation value, thereby enlarging the computing margin. In addition, the ADC is gated to lower power. Third, a dual degree of freedom array peripheral flexibly activates data from the search region and supports parallel computing and loading.

IV. Hybrid XOR-Derived CIM

This section details that CV-CIM integrates various distance functions based on the property of XOR logic and introduces the functionality of XOR-derived CIM, including pixel- and vector-wise XOR-CIM, respectively.

A. XOR *Derived Function*

To adapt to a diverse range of distance functions, XOR has been selected as the foundational logic by the CV-CIM. This decision is based on the inherent suitability of XOR operations for estimating distances between pixels, such as Hamming distance. Moreover, XOR logic proves that it can derive a multitude of additional operations, as illustrated in Fig. 5. For instance, fixing one of the inputs to 1 effectively transforms the XOR gate into an inverter.

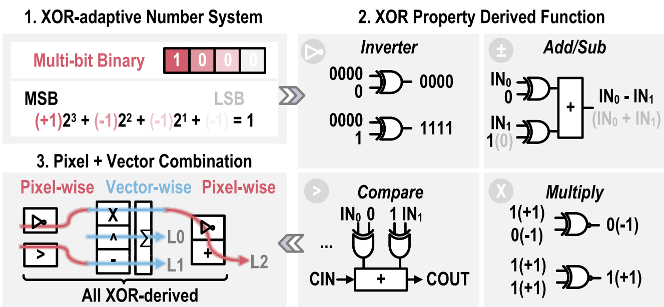

Fig. 5. Multi-bit binary representation and derived functions.

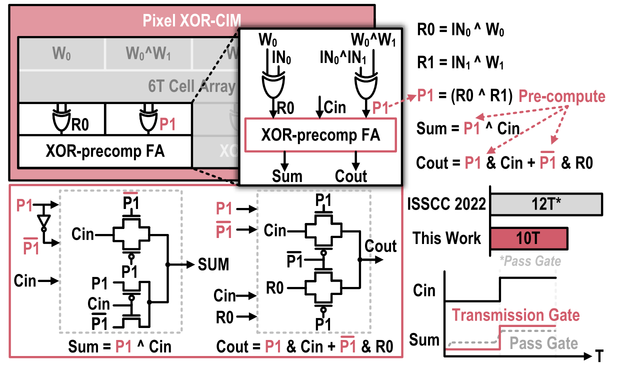

Fig. 6. Pixel CIM scheme and FA design.

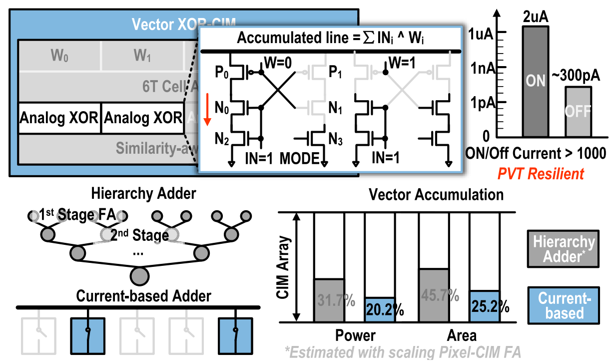

Fig. 7. Vector CIM scheme and accumulation comparison.

To further expand the realizable functions, CV-CIM leverages a number system that exhibits exceptional compatibility with XOR operations: multi-bit binary number representation (MBR). In this representation, each bit equals ± 1 and maintains bit significance. Each MBR value can be uniquely mapped to a corresponding value in 2's complement representation without any loss of accuracy. For instance, the MBR value "1000" equates to $(+1)2^3 + (-1)2^2 + (-1)2^1 + (-1)2^0$. Compared with 2's complement, MBR facilitates the conversion of positive numbers to their negative counterparts by a simple inverter. This advantageous property is exploited to accomplish pixel-wise addition, subtraction, and comparison operations by harnessing an inverter derived from XOR logic. In addition, MBR transforms certain necessary operations, such as MAC, into XOR and accumulation (XAC) operations. The primary overhead associated with MBR is the data format conversion. Considering few number of operations, the transformation from 2's complement to binary representation can be performed off-chip.

The distance algorithms are viewed as a combination of different basic functions. For example, the L1 distance computes the absolute difference between pixels, consisting of pixel-wise comparison, pixel-wise subtraction, and vector-wise reduction. To accommodate computations across varying dimensions, the hybrid XOR-CIM architecture offers support for both pixel- and vector-wise XOR operations, by pixel CIM and vector CIM, respectively.

B. Pixel XOR-*CIM*

Within the hybrid CIM group, the pixel XOR-CIM performs pixel-wise processing, yielding output dimensions that are the same as the input data. In computing mode, the data are read out from the cell and perform XOR with input within digital gates. An FA then merges the results from adjacent columns. To mitigate the area overhead associated with FAs, an innovative XOR-precompute FA is proposed.

Conventional FAs, reliant on complete CMOS logic, require 28 transistors to achieve SUM and carry-out (COUT) computations [35]. While pass gate adders offer a reduction in transistor count to 12, they introduce the undesired consequence of voltage drop [36]. The output result has a threshold voltage gap to VDD or GND, which may deviate the result once accumulated in multiple stages. This work achieves the rail-to-rail logic based on the transmission gate. The idea is to complete partial logic of the adder by the digital XOR gate. As shown in Fig. 6, the FA is designed to perform the summation of $\mathrm{IN}_0 \oplus W_0$ and $\mathrm{IN}_1 \oplus W_1$. It is observed that both the sum (SUM) and COUT logic share a common subexpression, denoted as **P1**: $\mathrm{IN}_0 \oplus W_0 \oplus \mathrm{IN}_1 \oplus W_1$. To optimize transistor count and maintain signal integrity, a digital XOR gate is employed for the pre-computation of P1. As shown in Fig. 6, one XOR gate calculates $\mathrm{IN}_0 \oplus W_0$, while the other stores the pre-computed result of $W_0 \oplus W_1$ within the bit cell, instead of storing W_1 individually. Subsequently, $\mathrm{IN}_0 \oplus \mathrm{IN}_1$ is used as the input to the CIM. As a result, the digital computation unit directly outputs P1, which is then utilized to compute both the SUM and COUT logic within the FA. This FA utilizes transmission gates to perform rail-to-rail XOR, AND, and OR operations. By reusing the pre-computed P1, the transistor count of the FA is reduced to 10 while ensuring signal integrity.

C. Vector XOR-*CIM*

Pixel-wise processing is delegated to the pixel CIM, producing results that independently serve as inputs for the vector CIM. This latter component executes vector computations and partial result reduction within a current-based paradigm. As shown in Fig. 7, the analog XOR unit, input data, controls the gate of N0/N2/P1 and weight in the cell decides the ON/OFF of P0/N1. The state of footer N3 depends on the mode signal.

In the original mode (mode $=$ 1), the footer transistor is activated. This allows current conduction through one of the designated paths when the input differs from the weight.

TABLE II
COMBINATION OF PIXEL/VECTOR CIM

Distance	Pixel CIM		Vector CIM		XOR Property
	Operation	Function	Operation	Function	
L0	-	-	$\text{Out}_v = \sum \text{IN}\oplus\text{W}$	XAC	Bit-wise Similarity
L1	Out_p = IN>W	Comparator	$\text{Out}_v = \sum \text{Out}_p\oplus\text{W}$	SUB,XAC	Self-inverse
L2	Out_p = !IN	Inverter	$\text{Out}_v = \sum \text{Out}_p\cdot\text{W}$	MAC	Bit flipping, Binary-Multiply

For instance, if IN equals 1 and W equals 0, the NMOS transistors N0 and N2, along with the PMOS transistor P0, become conducting, establishing a current path on the left side. This current contributes to the horizontal accumulation line (AL). Conversely, when the input matches the weight, at least one transmission gate in both paths is turned off, resulting in no current contribution to the AL. In contrast to prior designs utilizing multi-state analog computing units [5], this design simplifies the computation by limiting the output to either 0 or 1 (corresponding to a conductive or non-conductive current path). The conducting path attains a current of 2 μA, while the cutoff path exhibits a current of 300 pA, thus achieving an ON/OFF ratio exceeding 10^3. The ratio guarantees the linearity under PVT variation. All partial results are merged in AL, which is subsequently quantized by a similarity-aware ADC. Albeit incurring ADC, the vector CIM design mitigates the challenge of pitch matching with a chunk adder tree. Compared with the adder tree scheme, the current-based accumulation reduces the area overhead from 45.7% to 25.5%. The cutoff path has no current contributions, which also reduces the power consumption from 31.7% to 20.2%.

By combining the pixel and vector CIMs, the hybrid CIM group can compute L0/L1/L2 distance, which is detailed in Table II. Pixel-wise computing, such as element comparison within the L1 distance calculation, is completed by the pixel CIM. The results are then shared by vector CIM as inputs. The vector CIM reduces vector-wise results, such as the XAC in L0 distance. CV-CIM also takes advantage of the connection between the L2 distance and the dot product [20]. The L2 distance between normalized pixels can be computed by the multiplication and accumulation. This design employs more vector CIM than pixel CIM due to its dominance in workload-specific vector operations. These operations involve accumulating pixel-wise data produced by the pixel CIM unit. To amortize the pre-processing overhead, eight vector CIMs share the result produced by the pixel CIM. Harnessing the versatile properties of XOR logic, all required functions can be derived from basic XOR operations. By reconfiguring the functionality of its computing units, CV-CIM improves the area density by 2.3× than digital distance units with identical throughput (0.36-mm^2 CV-CIM versus 0.26-mm^2 SRAM + 0.054-mm^2 L0 distance [4] + 0.043-mm^2 L1 distance [22] + 0.486-mm^2 L2 distance [37]).

V. SIMILARITY-AWARE XOR TRANSFORM

This section introduces the theoretical analysis of the compute margin and margin optimization based on similarity.

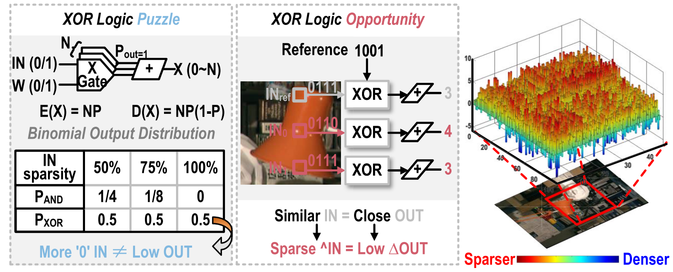

Fig. 8. XOR-derived puzzle and opportunity.

A. CIM Compute Margin and XOR-Derived Puzzle/Opportunity

For CIM design, the concept of the computing margin emerges as a critical metric for accurate quantization. This margin quantifies the interval between neighboring output values. In ideal scenarios where these intervals are uniform, the compute margin can be approximated as the ratio of the feasible physical quantity range to the maximum accumulated output. As the physical quantity commonly remains fixed, the margin is inversely proportional to the accumulated output. The output generated by the CA, a product of merging the computing results of input and weight pairs, adheres to a binomial distribution. Both the expectation and the variance of the output range forge a close association with the number of input pairs (N) and the probability of the output assuming a value of 1, denoted as $P(\text{output} = 1)$. The conventional AI CIM employs AND as basic logic for executing convolution and FC layers. The $P(\text{output} = 1)$ of AND logic reduces with higher data sparsity as the output equals 0 once the input is 0. In contrast, $P(\text{output} = 1)$ of XOR does not reduce with input sparsity, thus inducing a wider output range and limited compute margin.

Despite the challenges associated with XOR logic, an opportunity has been noticed that computations involving similar inputs yield comparably similar outputs, resulting in relatively low Δoutput. This property can be leveraged to mitigate the reduced margin issue. Rather than directly computing the absolute values of inputs and weights, the CIM processes the differences between consecutive inputs. This approach produces a computational outcome that represents a deviation from a fixed offset, typically owning a lower magnitude than the original output. To exploit this opportunity, a mathematical transformation is introduced, converting the XOR operation between original inputs into an AND logic operation between the difference and reference, as illustrated by (1).

Under the assumption that adjacent pixels have a significant degree of similarity, the similarity result is close to 0, which lowers the $P(\text{output} = 1)$ of AND logic and contributes to a further reduction in the accumulation range. The efficacy of this approach is visually depicted in Fig. 8, which shows the change in the number of bit 0s after computing the difference between a reference pixel and its neighboring pixels. Red polar coordinates denote pixels that have become sparser following the XOR operation with the reference pixel, while the blue polar coordinates mean that the generated result becomes dense instead. Overall, the ratio of bit 0s within each pixel

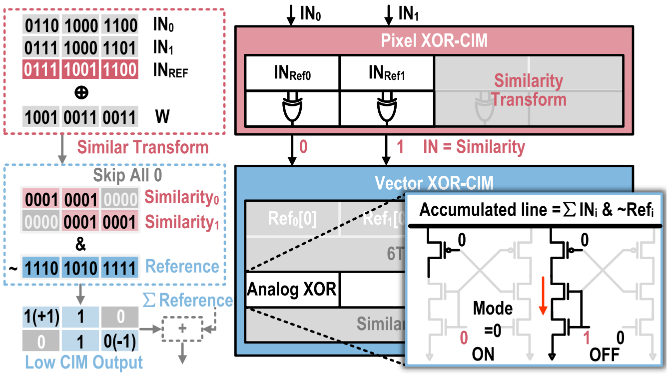

Fig. 9. Similarity transform and sparse-mode CIM.

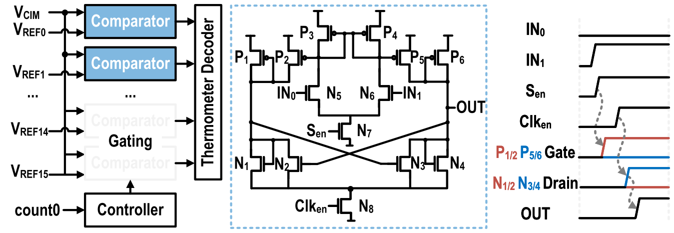

Fig. 10. Similarity-aware ADC and comparator design.

demonstrates an average increase of 27%

$$\begin{aligned}
&\sum \mathrm{IN}_i \oplus W_i \\
&\quad = \sum \mathrm{IN}_i \oplus \mathrm{IN}_{\mathrm{ref}} \oplus \mathrm{IN}_{\mathrm{ref}} \oplus W_i \\
&\quad = \sum (\mathrm{IN}_i \oplus \mathrm{IN}_{\mathrm{ref}})\ \&\ \overline{\mathrm{IN}_{\mathrm{ref}} \oplus W_i}) + \sum (\mathrm{IN}_{\mathrm{ref}} \oplus W_i) \\
&\quad = \sum \mathrm{Difference}\ \&\ \overline{\mathrm{Reference}} + \sum \mathrm{Reference} \qquad (1)
\end{aligned}$$

where Difference equals $\mathrm{IN}_i \oplus \mathrm{IN}_{\mathrm{ref}}$ and Reference represents $\mathrm{IN}_{\mathrm{ref}} \oplus W_i$.

B. Sparse-Mode CIM

Fig. 9 depicts the implementation of similarity-aware optimization. This technique leverages data similarity to augment compute margins, which employs the block center pixel as a reference input ($\mathrm{IN}_{\mathrm{ref}}$) for comparison with subsequent inputs within the same block. The similarity transform step generates a sparser difference result, consequently reducing accumulation and enlarging margins. However, this optimization incurs additional operation overhead, including block-wise "comparisons" with the reference pixel and "AND" operations between generated data similarity and the reference.

Hybrid XOR-CIMs are employed to efficiently execute these operations. The pixel CIM component conducts the input similarity transform, and its overhead is amortized by sharing similarity across eight parallel vector CIMs. Furthermore, all-0 inputs are detected and bypassed to optimize the performance. The XOR-based accumulation is then transformed into an AND between similarity and inverted reference. Vector CIM operates in sparse mode (mode = 0) to execute the operation IN & ∼W (bitwise AND of input and complement of weight). In this mode (mode = 0), the footer transistor of the right path is gated, effectively cutting off this path. Consequently, only the left path can contribute current to the horizontal AL. However, current conduction through this path is only possible when the specific combination of IN = 1 and W = 0 occurs, allowing all transistors in the path to be turned on. The number of occurrences of this specific combination is accumulated and quantized by the flash ADC.

C. Similarity-Aware ADC

The similarity-aware optimization effectively reduces accumulation results, enabling a lower precision quantization strategy. Analysis of output distribution demonstrates that a 4-bit quantization adequately covers over 99.3% of outputs. Besides, cost–volume construction primarily focuses on candidates with low values, further supporting the adoption of 4-bit ADCs.

Observing the reduced output range, a similarity-aware ADC design saves power by selectively activating comparators because not all comparators are essential when CIM results fall significantly below 16. To exploit this opportunity, the ADC monitors input data and counts non-zero elements. When half of the inputs are zero, the controller periodically deactivates comparators of the corresponding high-voltage ladder to reduce power. The technique helps reduce an average of 9.90% of power consumption in distance computing.

Fig. 10 presents a detailed schematic of the comparator circuit designed to differentiate voltages ranging from 0 to VDD. This differentiation process is accelerated by a multi-stage comparator. The first stage amplifier, comprised of transistors P3, P4, N5, and N6, becomes active upon receiving the sense enable signal (sen). In this activated state, the amplifier begins the amplification of the differential voltage between IN0 and IN1. The amplified output from the first stage is fed into a cross-coupled latch formed by transistors P1, P2, P5, P6, N1, N2, N3, and N4. The amplified differential voltage from the first stage also creates a potential difference at the drains of transistor pairs N1/N2 and N3/N4. This latch mechanism expedites the voltage differentiation process upon the arrival of the clock enable signal ($\mathrm{Clk}_{\mathrm{en}}$). For instance, if the voltage at the drain of N1/N2 is higher than that of N3/N4, the latch will completely turn on the right path, driving the drain of N3/N4 to the ground while forcing the drain of N1/N2 to VDD. This regenerative action by the latch circuit significantly enhances the speed of voltage differentiation.

VI. Dual-Degree-of-Freedom Peripheral

This section explains how CV-CIM improves the utilization during the cost–volume construction in two dimensions, column-wise and row-wise.

A. Column-Wise Peripheral

In conventional CIM architectures, the search plane's horizontal shifting during search iterations induces data loading, often leading to performance degradation due to compute stalling. This is exacerbated by memory bandwidth limitations, which fall significantly below the parallelism of CAs times array column numbers. Consequently, multiple cycles are required to refresh the entire CIM group. Existing approaches,

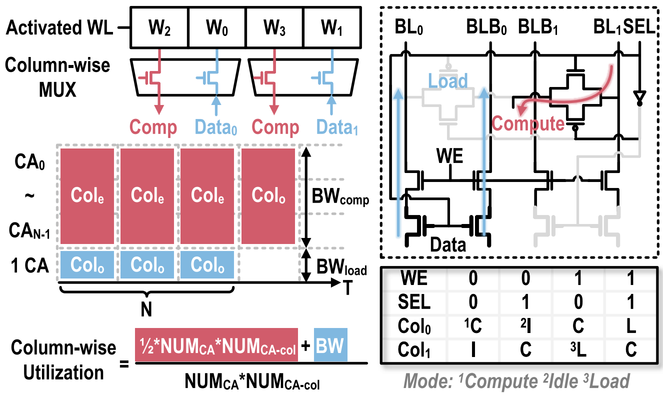

Fig. 11. Column peripheral design and pipeline optimization.

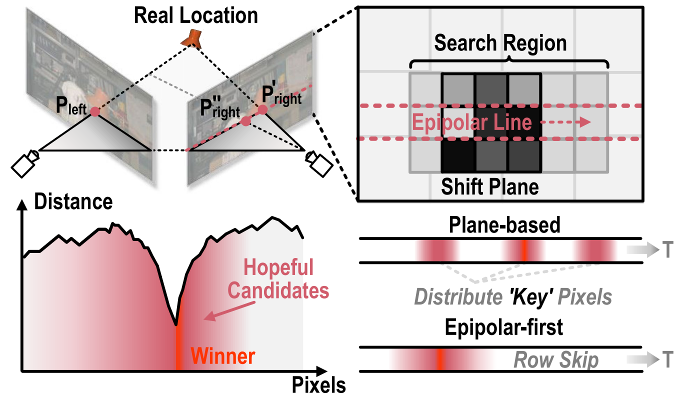

Fig. 12. Epipolar-first search optimization.

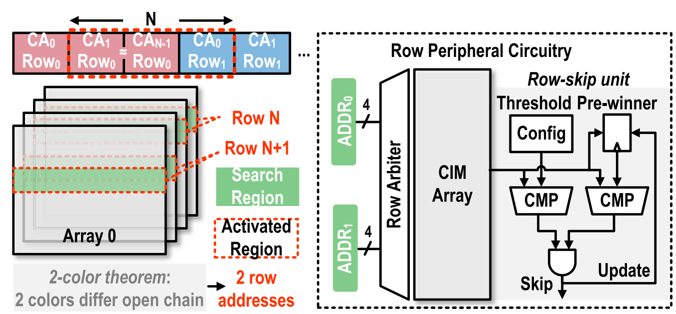

Fig. 13. Row peripheral optimization and circuit design.

such as ping-pong CIM [38], enable concurrent computation and weight updates. This design utilizes two data banks that share a single local computing unit. While one bank performs computations, the other can be loaded with new weights. However, this technique necessitates duplicating the write-assist circuitry, resulting in a significant area overhead. In addition, activating two WLs for computation and loading increases power consumption. To address this bottleneck, CV-CIM introduces a column peripheral that enables parallel writing and computing, as dedicated in Fig. 11. The peripheral employs a column-wise MUX, decided by the SEL signal, to engage half of the columns for computing while simultaneously enabling data writing to the complementary columns upon same WL activation. With column-wise freedom, the idled columns occupied by writing are released for computing.

The computing bandwidth is N times higher than the write bandwidth, where N represents the number of CAs. When weight reuse exceeds N, the data loading latency can be entirely hidden through pipelining. N equals 36 in CV-CIM, consisting of 32 vector CAs and four pixel CAs. Fig. 11 exemplifies this mechanism, with red columns representing computing operations and blue columns denoting loading activities. Even columns (Col_e) across all CAs execute computations, while one CA simultaneously writes data to odd columns (Col_o). After N cycles, the odd columns of all CAs become available for subsequent computations. To minimize area overhead, the column MUX leverages the intrinsic circuitry of the write driver, resulting in a negligible overhead of less than 0.57% without compromising pitch match. Moreover, compute and load columns share the SEL and its complement signal, promoting efficient resource utilization.

B. Row-Wise Peripheral

The row-wise peripheral optimizes search behavior and facilitates flexible data activation. While the global search for matching pixels poses an NP-hard challenge, geometric constraints provide a potential optimization. Specifically, geometric principles dictate that potential matching pixels reside along a horizontal epipolar line, as depicted in Fig. 12. Conventional search methods often adopt a plane-based accumulation approach. It exhaustively searches for matching pixels within a defined block (as exemplified by the black box in Fig. 12). Once a block is processed, the algorithm iterates to neighboring blocks until the entire search region is covered. However, this strategy scatters promising candidate pixels that originate from the epipolar line. CV-CIM proposes an innovative epipolar-first search scheme that prioritizes the computation of candidate matches solely on the epipolar line. This prioritization enables selective skipping of non-significant pixels based on comparisons with a pre-iteration winner and a pre-tuned threshold. The pre-iteration winner assumes that neighboring pixels tend to have similar depths. Therefore, the winner from the previous iteration serves as a reference point. By considering only the pre-defined threshold, the technique reduces computations by approximately 19% while maintaining an accuracy loss of 12.0%. The pre-winner approach reduces 26% operations with 19.9% accuracy loss. When satisfying both comparison requirements, search skipping yields an 11% operations reduction with an accuracy loss of less than 2.3%.

Beyond selective skipping, the epipolar-first search paradigm offers the additional benefit of dimension reduction, effectively transforming the search process into a 1-D shifting operation. The data mapping also adapts to the search process. Sequentially, neighboring pixels are stored within the same row of distinct CAs (row0 of CIM array0, row0 of CIM array1, ...) and progress to subsequent rows when all arrays are occupied (row1 of array 0, row1 of array1, ...). Conceptually, the mapping resembles an open chain, through which an N-width search region slides, where N denotes the number of arrays. Harnessing the principles of the two-color theorem, CV-CIM's row-wise peripheral requires only two addresses to accommodate all sliding steps. As detailed in Fig. 13, the row arbiter selects two addresses, activating the

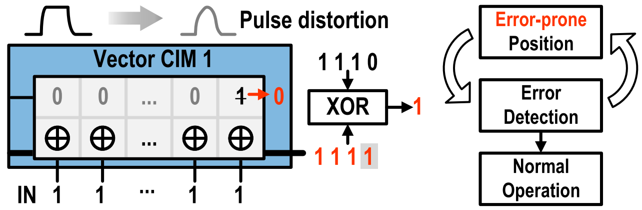

Fig. 14. Design and workflow of canary-style BIST unit.

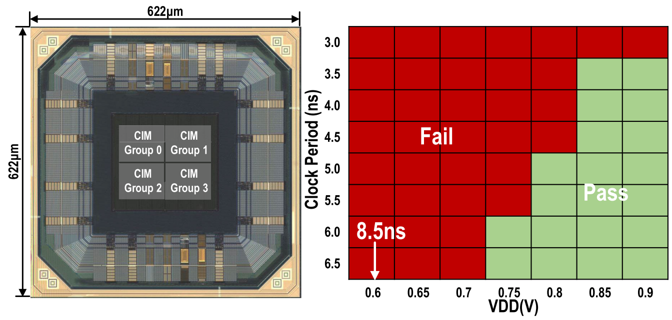

Fig. 15. Die photograph and shmoo plot results.

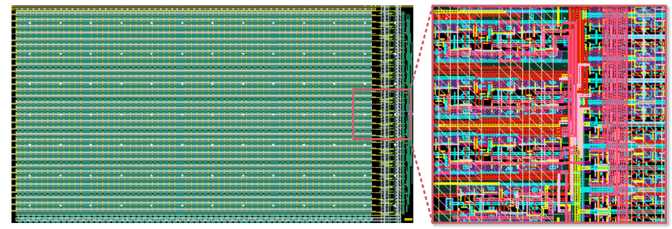

Fig. 16. Layout of CV-CIM bank array and peripheral.

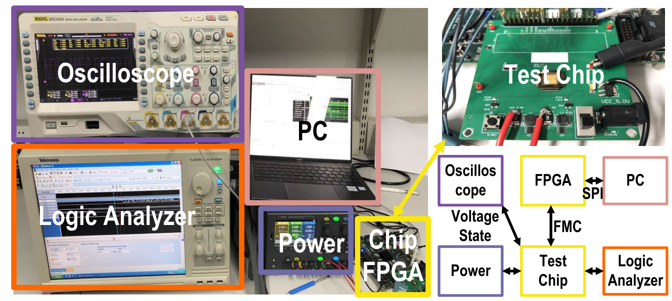

Fig. 17. Verification system.

corresponding array. The resulting output is then compared with both the pre-determined winner and a pre-tuned threshold to determine whether row skipping is permissible. Eventually, the value of the pre-winner register is updated for the following skipping comparison.

VII. Chip Prototype

A. Canary-Style BIST

To mitigate the susceptibility of analog vector CIM units to PVT variations, CV-CIM integrates a canary-style BIST mechanism inspired by prior work [39], as shown in Fig. 14. The principle of this proactive approach is failing before normal computations. The BIST unit operates by replicating a worst case scenario and triggers fail-early mechanisms once detecting faults. The design strategically targets the farthest column within the CA from the WL pulse generator. This specific location experiences the most severe signal distortions due to its extended distance from the WL driver. The BIST unit stores a predetermined bit value within this column and subsequently verifies the computed output against a known golden result. A comparison output of 0 indicates a successful match, whereas a value of 1 signifies instability within the column. Upon detection of a faulty column, the BIST unit then discards it and initiates testing on the next rightmost column within the remaining array. This iterative testing process continues until all detected columns produce the correct results.

B. Chip Prototype

The CV-CIM is fabricated in 28-nm TSMC CMOS technology. Fig. 15 shows the die photograph. The chip area is 0.622 × 0.622 mm, which contains the 36-kb CIM group, along with essential digital peripherals and I/O pads. Each CIM group comprises eight vector CIMs and one pixel CIM. Vector CIM and pixel CIM occupy the minimal areas of 0.0015 and 0.0011 mm^2, respectively. The CV-CIM demonstrates functioning within a voltage range of 0.6–0.9 V and a frequency range of 50–286 MHz. At 0.6 V, the chip achieves a peak energy efficiency of 1158 TOPs/W. Fig. 16 shows the layout of the CV-CIM bank array and row-wise peripheral.

Fig. 17 details the verification system for CV-CIM. A dedicated dc power supply provides CV-CIM with various voltage levels, including I/O and core voltages. Core voltage is further segmented into memory array, array peripheral, digital unit, and ADC reference voltages. Acting as the host processor, a Xilinx Zedboard field-programmable gate array (FPGA) board feeds the CV-CIM with test image data from the KITTI2012 dataset. The dataset provides 1240 × 376 pixel images for performance evaluation. To comprehensively assess the CV-CIM's efficiency and throughput across diverse algorithms, L0/L1/L2 distance distances are independently evaluated. These distances are employed by algorithms such as census transform, sum and absolute difference, and MC-CNN. A logic analyzer captures the CV-CIM's results and compares them against expected values, ensuring the correctness of computations.

VIII. Measurement Results

A. CIM Linearity

The vector CIM employs current-based accumulation, and the ideal output remains stable and distinguishable even with the increasing of accumulating value. However, as more contributing paths feed current into the AL, transistors within the local computing unit begin to behave non-linearly, degrading expected functionality. This deviation results in reduced compute margins and potentially introduces quantization errors. To maintain accuracy and relieve the burden of the flash ADC, the maximum activated number of columns of each array is constrained to 32. This mitigates the negative impact of non-linear transistor behavior on output stability. To quantify the effects of PVT variations on linearity, dedicated experiments were conducted.

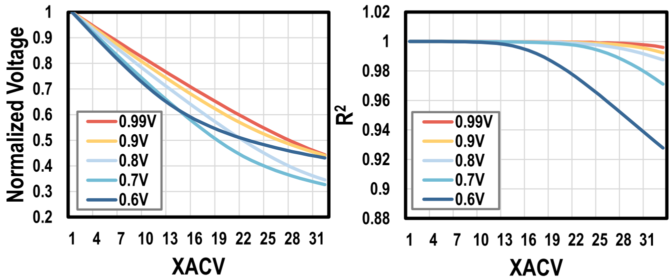

Fig. 18. Linearity of CV-CIM and R^2 experiments.

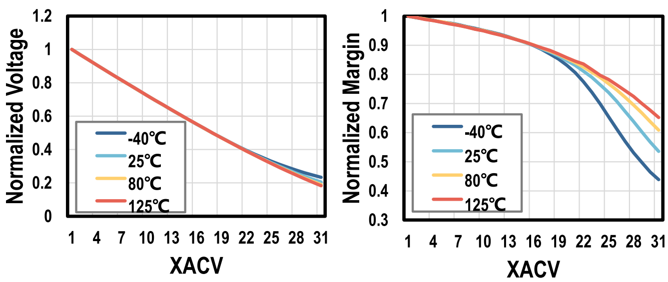

Fig. 19. Temperature effect on linearity and compute margin.

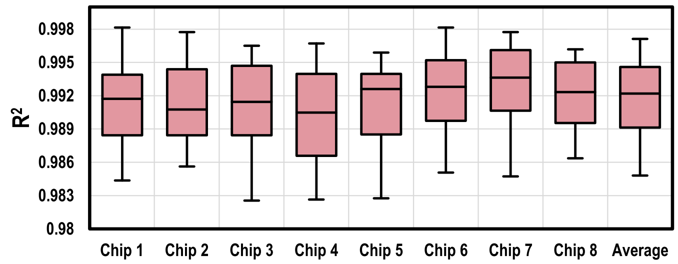

Fig. 20. Fabrication effect on computing linearity.

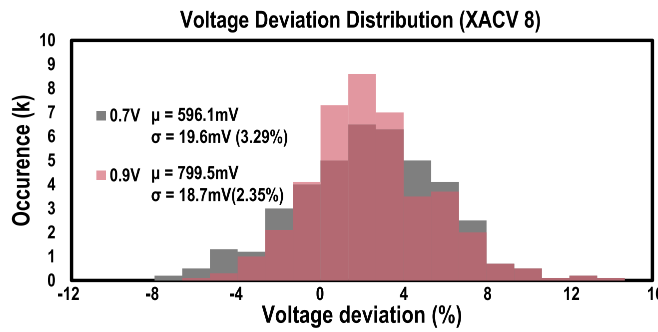

Fig. 21. Voltage deviation distribution of CV-CIM with XACV 8.

Fig. 18 depicts the relationship between the voltage on the AL and the XOR accumulated value (XACV) under different supply voltages. The voltage of the y-axis is normalized to the initial voltage of XACV 1. Above 0.8-V supply voltage, the XACV curve exhibits excellent linearity across all values. At 0.6 V, deviations appear from XACV 17 onward. To evaluate the accuracy performance of the computing unit, the coefficient of determination (CoD) serves as a quantitative measure of this deviation, representing the squared correlation coefficient (R^2) between predicted and actual performances. R^2 of 1 means that the output exactly matches the ideal linearity curve. As illustrated in Fig. 18, the CoD remains above 0.99 for supply voltages between 0.8 and 0.99 V. At 0.7 V, it holds steady above 0.96. Even at 0.6 V, the CoD is above 0.98 when the XACV is below 16.

Fig. 19 delves into the impact of temperature on CV-CIM's linearity. Under a consistent 0.9-V supply voltage, the analysis assesses linearity across various temperature nodes. Similar to Fig. 18, the AL voltage is normalized to XACV 1. Before exceeding an XACV value of 22, CV-CIM exhibits exceptional linearity across all temperature conditions. The R^2 metric surpasses 0.99 in every case. For XACV values above 22, the temperature begins to affect the linearity performance, and higher temperatures demonstrate superior linearity compared to lower temperatures. However, even in the worst scenario of −40 °C, CV-CIM retains an acceptable CoD of 0.95.

To further quantify the linear performance, the interval voltage between neighboring nodes is calculated, also referred to as the computing margin. A thorough examination of the voltage intervals between all neighboring values is conducted, with the results subsequently normalized to their initial values. Unlike the simplified model, these voltage intervals are not identical and shrink as the accumulated value increases, as the transistors operate outside the linear region when the accumulated voltage on the horizontal line overpasses the threshold. At room temperature, the margin at the highest XACV retains 54% of its original value. At 125 °C, the margin remains above 67%.

To validate the CV-CIM's linearity under manufacturing process variations, we conducted tests on eight chips fabricated using TSMC 28-nm CMOS technology. Under a constant supply voltage of 0.9 V, the performance was evaluated for XACV values ranging from 1 to 31. Fig. 20 presents the average R^2 and the minimum and maximum R^2 values across all chips and XACV levels in the box plot. While minor performance deviations are observed, the average R^2 remained consistently above 0.990, proving robust linearity. Even the worst performing chip 3 achieves an R^2 of 0.983. The average performance of all chips spanned from 0.984 to 0.997, demonstrating the effectiveness in mitigating process variations.

Also, Monte Carlo simulations are conducted to prove the computing reliability of CV-CIM at different voltage nodes. Fig. 21 reveals the voltage distribution observed when the XAC value equals 8. A lower voltage deviation represents a concentrated distribution of computing results and enhances the quantization accuracy. At 0.7-V supply voltage, the voltage deviation is less than 3.29%, and the ratio improves to 2.35% when increasing the supply voltage to 0.9 V.

These extensive tests demonstrate the CV-CIM's computing robustness across a wide range of PVT variations.

B. Computing Margin Optimization

Based on the findings that the CIM margin degrades with a higher accumulation value, Section V introduces a similarity-aware transform to effectively mitigate this challenge. This optimization relies on pixel similarity to generate

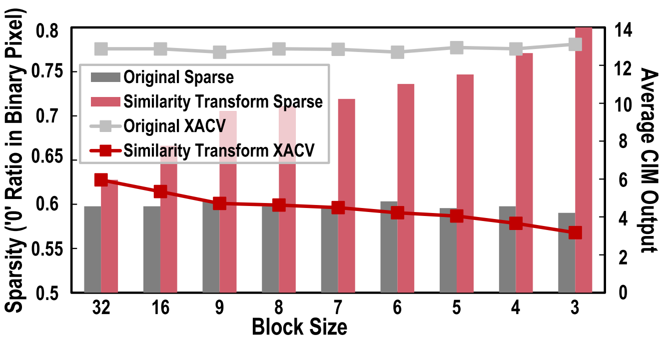

Fig. 22. Compute margin of CV-CIM on different block sizes.

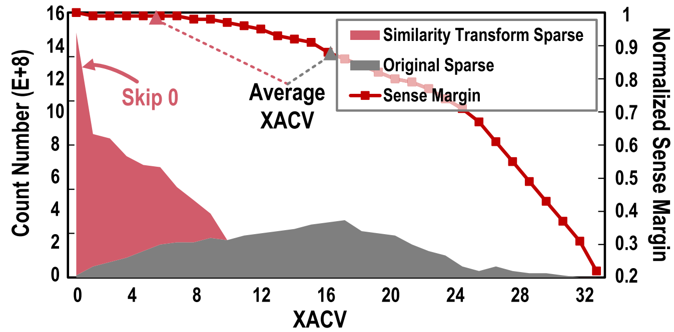

Fig. 23. Occurrence of the XACV and corresponding compute margin.

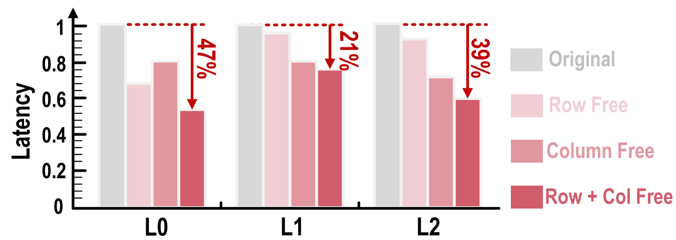

Fig. 24. Latency optimization with dual-degree-of-freedom peripheral.

bit-wise sparsity. To achieve this, blocks of input pixels are compared with a reference pixel. Fig. 22 presents the sparsity ratio after performing similarity transform in different block sizes. The sparsity metric is the ratio of "0" bits within the total bits. Specifically, even a block size of 9 exhibits a substantial 17% sparsity increase compared to scenarios without the transform. This enhancement even increases with smaller block sizes, reaching a remarkable 36% improvement for a block size of 3. This phenomenon is because of the inherent continuity of adjacent data and the resulting contextual similarity, leading to sparser outcomes when neighboring values are compared.

The increasing sparsity directly translates to reduced CIM output values, as illustrated by (1). By reformulating XOR operations into AND operations, the transform can utilize data sparsity to effectively lower accumulation results. The findings reveal a significant sparsity enhancement ranging from 55.2% to 75.7%. Specifically, the average output without the transform is 12.9, which reduces to 5.7 with the transform within a block size of 9. Further downsizing the block size to 3 yields an output reduction to an average of 3.6.

Fig. 23 analyzes the impact of the similarity-aware transform by examining the distribution of XACV occurrences within the KITTI2012 dataset. This figure demonstrates that the distribution of the accumulation output occurrence is concentrated within a lower value range. Moreover, the ability to bypass all-zero cases during computation contributes to a 5.2% reduction in latency. The occurrence presents that similarity-aware transform reduces accumulated output by 4.13×. Next, the corresponding compute margin of XACV can be acquired from earlier experiments. The average margin ascends from 84.1% to a remarkable 98.7% after similarity optimization.

C. Dual-Degree of Peripheral Optimization

To optimize the utilization of CIM, a dual-degree-of-freedom peripheral is integrated within CV-CIM. This innovative approach involves two components, row-wise peripheral and column-wise peripheral. The row-wise peripheral flexibly activates data within the search region and selectively skips redundant computations. The column-wise peripheral enables parallel data updating and computing.

Fig. 24 demonstrates the latency reduction achieved by this dual-degree optimization. The row-wise peripheral demonstrates a conservative approach, and skipping should satisfy both pre-winner and threshold comparisons, leading to latency improvement of merely 4.1% in L1/L2 distance computations. However, the flexible row peripheral effectively activates required rows to avoid coalescing search regions in multicycle, which reduces L0 computing latency by 35.6%. The column-wise peripheral proactively prepares data during computing. The analysis of the KITTI2012 benchmark dataset reflects the distinct data reuse patterns across these algorithms. L0 distance computation achieves an average data reuse of 19.734×, L1 distance achieves 7.612×, and L2 distance achieves 12.035×. While the pipeline may not entirely eliminate update latency in all cases, the column peripheral architecture efficiently reduces it for each application. The magnitude of latency reduction is influenced by both data reuse and weight loading frequency. The experimental results demonstrate that the parallel update and compute capabilities facilitated by the column peripheral contribute to an average latency reduction of 20.1%, 20.7%, and 26.1% for L0, L1, and L2 distances, respectively.

These two optimizations operate orthogonally. By harnessing the complementary strengths of row- and column-wise peripherals, CV-CIM achieves a remarkable latency reduction of 21%–47% across L0, L1, and L2 distance algorithms.

D. Hardware Efficiency

Table III comprehensively showcases the hardware performance of CV-CIM across diverse algorithms. To assess its efficiency, L0/L1/L2 distances are evaluated independently.

For L0 distance computing, fully activated computing units achieve an exceptional throughput of 1259.2 TOPs. This peak performance coincides with the best energy and area efficiencies at 0.6 V, reaching 1158 TOPs/W and 27 TOPs/mm^2. SAD is the simplest algorithm used in stereo vision. It calculates disparity within a limited region to generate the output disparity map. This simplicity translates to fewer operations and

TABLE III
COMPUTING-IN-MEMORY EFFICIENCY

	L0				L1				L2		
Algorithm[1 2]	Census Transform[17]				SAD[18]				Algorithm	MC-CNN[21]	
Disparity Size	64		128		64		128		Type	Concate	Correlation
Plane Size	9*9	11*11	9*9	11*11	7*7	9*9	7*7	9*9	Parameter(M)	0.6	8.75
Weight Precision	8	8	8	8	4	4	4	4	Weight	8	8
Activation Precision	8	8	8	8	4	4	4	4	Activation	8	8
Operation(G)[3]	4.834	7.222	9.666	14.442	2.924	4.834	5.848	9.666	Operation(G)	20.82	38.56
Accuracy[4]	87.38 (85.46)	89.41 (88.17)	91.31 (90.51)	92.73 (91.65)	78.77 (71.11)	80.54 (74.44)	81.22 (79.15)	85.75 (81.97)	Accuracy	97.39 (96.16)	97.84 (97.29)
Utilization(%)	79.0	94.5	79.0	94.5	43.7	56.25	43.7	56.25	Utilization(%)	88.0	91.12
Latency(ms)	4.49	5.74	9.34	12.00	4.72	7.05	9.60	14.71	Latency(ms)	17.4	31.5
Throughput(GOps)[5]	1074.82	1259.24	1035.4	1204.24	619.74	685.98	601.23	658.22	GOPs	1193.82	1224.4
Energy Efficiency(TOPs/W)	969.6121	1158.281	946.935	1193.062	471.0449	505.501	453.2223	485.7179	TOPs/W	988.6942	1092.168
Area Efficiency(TOPs/mm2)	23.36565	27.37478	22.5087	26.17913	13.89783	14.91261	13.18826	14.30913	TOPs/mm2	25.95261	26.61739
Energy/Operation(fj)	1.03134	0.863348	1.056039	0.838179	2.060395	1.978235	2.104701	2.058808	fj	1.011435	0.91561
Energy/Search(uj)	4.985499	6.235102	10.20767	12.10499	6.100595	9.56279	12.460292	19.90044	uj	21.05808	35.30592

[1] Kitti2012 1240*376 resolution image
[2] Estimate the matching distance only, ignore post-processing
[3] Distance + Add (1b) is denoted as 1 op
[4] Non-occluded(total)
[5] Normalized to 1b, including all-0 skip & Row skip.

a smaller memory footprint, resulting in an energy efficiency range of 453∼505 TOPs/W and an area efficiency range of 13.1∼14.9 TOPs/mm^2. The limited data reuse and computing parallelism hinder its ability to fully leverage the advantages of CIM architectures. MC-CNN is adopted to evaluate the performance of L2 distance and CV-CIM realizes an energy efficiency of 1092 TOPs/W and 26.6 TOPs/mm^2. L0 distance computing consumes 4.49∼12.0 ms depending on the search region size. While limited by utilization, L1 distance computing only takes 4.72∼14.71 ms due to fewer required operations. L2 distance computing is slowest and consumes up to 31.5 ms, but the latency penalty is offset by accuracy, with 97.84% in non-occluded scenarios, and maintains 97.29% in overall cases.

Based on previous findings regarding CV-CIM's operational range (50 MHz at 0.6 V and 286 MHz at 0.9 V), Fig. 25 shows the impact of voltage scaling on its efficiency across various benchmarks. For energy efficiency, all benchmarks exhibit the optimal performance at the lowest voltage (0.6 V), with an average peak of 918.6 TOPs/W. As the supply voltage increases to 0.9 V, a decline in energy efficiency is observed, with the average dropping to 578.5 TOPs/W. Area efficiency follows a similar pattern, reaching its best performance of 22.9 TOPs/mm^2 at 0.6 V and reducing to 16.0 TOPs/mm^2 at 0.9 V.

Fig. 26 presents a detailed breakdown of the chip area and power. The CA is the main component of this chip, which contains cell array, row/column peripheral, XOR-derived computing unit, and FA/ADC. These components claim three-quarters of the chip's area and roughly two-thirds of its overall power consumption. The memory controller is compulsory for generating address and instruction, requiring

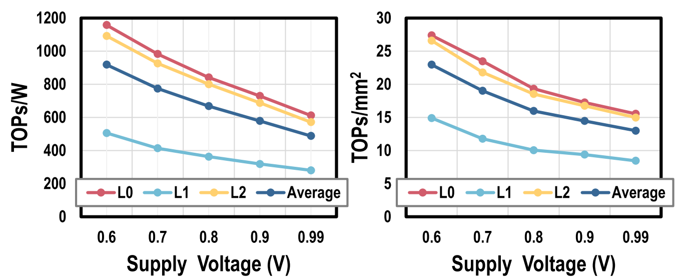

Fig. 25. Energy efficiency and area efficiency of CV-CIM.

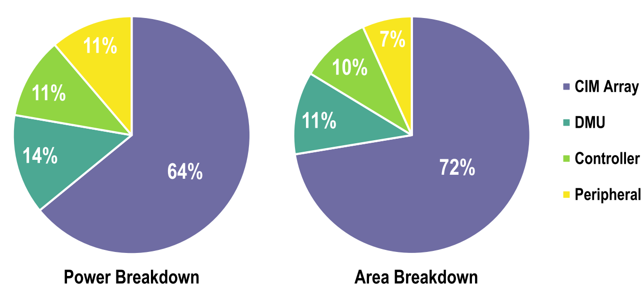

Fig. 26. Area and power breakdown.

approximately 10% of both area and power resources. Compared to an intrinsic memory array, CV-CIM incorporates a DMU to consolidate results from multiple arrays, which induces at most 11% area and 14% power overhead. In addition, other memory peripherals, such as temporal buffer, BIST unit, and configuration unit, collectively contribute an additional 7% to the area budget.

E. Comparison Results

Table IV shows the comparison with the recent CIM works [36], [40], [41], [42], [43], [44]. To ensure a fair

TABLE IV
COMPUTING-IN-MEMORY COMPARISON

	ISSCC2018[40]	JSSC2020[41]	JSSC2020[42]	JSSC2021[43]	ISSCC2022[44]	ISSCC2022[36]	**This Work**
Technology	65nm	65nm	65nm	65nm	28nm	28nm	**28nm**
CIM Type	Analog	Analog	Analog	Digital	Digital	Analog	**Analog + Digital**
CIM Function	XNOR	XNOR	XNOR	XNOR	XOR, MAC	XNOR	**XOR,CMP,ABS,MAC**
Power Supply	0.725	1.0	0.6∼1.0	0.6∼0.8	0.8	0.45∼1.1	**0.6∼0.9**
Frequency(MHz)	434	-	∼400	138	333	250/280	**50∼286**
Cell	Split 6T	8T1C	12T	6T+XNOR[2]	6T	6T	**6T**
Cell Area(um^2)	0.525	-	3.915	10.53	-	-	**0.307**
Array Area(mm^2)	-	0.081	0.1125	0.2272	0.03	0.033	**0.0011(Pixel) 0.0015(Vector)**[3]
Array Size	4Kb	16Kb	16Kb	16Kb	32Kb	16Kb	**36Kb**
IN/W Precision	1/1	1/1	1/1	1∼16/1∼16	1∼8/1∼8	1/1	**1∼8/1∼8**
Power Efficiency (TOPs/W)[1]	55.8	671.5	403	156	219.04	1108	**206∼1158**[4]
Area Efficiency (TOPs/mm^2)[1]	-	20.22	5.461	2.496	-	-	**4.67∼27.36**[5]

[1] scaling to 1b/1b activation/weight.
[2] incur an XNOR gate within the cell array
[3] layout area of one CIM array
[4] peak census L0 distance performance tested at 0.6V supply voltage
[5] consider CIM array area size, exclude I/O interface, PLL and other peripheral.

and relevant comparison, CV-CIM selects other recent CIM works utilizing XOR/XNOR logic and similar technology. Compared to prior works, CV-CIM integrates variable arithmetic functions within its XOR-derived logic. This functional expansion empowers it to accommodate the diverse computational demands of different distances. CV-CIM also optimizes the computing margin by taking advantage of pixel similarity. In addition, the sparsity is exploited to gate comparator and bypass redundant operations for greater gains. The flexible array peripheral also plays a pivotal role in improving CIM utilization, which selectively avoids unnecessary row searches and minimizes loading periods. Finally, CV-CIM leverages the intrinsic SRAM array and peripherals to lower area overhead. Compared to prior CIM works, CV-CIM achieves energy efficiency improvement ranging from 1.05× to 5.29× and area efficiency enhancements of 1.35∼5.01×.

IX. CONCLUSION

This work accelerates cost–volume construction by introducing CV-CIM, a novel in-memory computing architecture that overcomes the limitations of prior CIM designs, including restricted function, reduced margin, and rigid array. CV-CIM employs three key features. First, recognizing the need for functional versatility, CV-CIM integrates diverse arithmetic functions within its hybrid XOR-derived logic. Second, CV-CIM takes advantage of pixel similarity to reformulate the original operation to enlarge the computing margin. Third, a flexible array peripheral improves the CIM utilization in both column and row dimensions. Fabricated in 28-nm CMOS with a compact 0.387-mm^2 area, CV-CIM exceeds existing CIM by up to 5.29× in energy efficiency and 5.01× in area efficiency.

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